

Hydraulics (CE 247) Past Questions Analysis (2079 -

Unit-Wise Past Question Bank (2079 - 2082)

This section compiles every single question from the past four years (2079 Regular, 2079 Supplementary, 2080 Regular, 2081 Regular, 2081 Mid-Term, and 2082 Regular). No questions have been omitted.

PART A: Fluid Flow in Closed Conduits (40% Weightage)

Unit 1: Basics in Pipe Flow

Focuses on laminar/turbulent shear stress & velocity distributions, Darcy-Weisbach equation, Moody's diagram, minor losses, and HGL/TEL.

Short Answer Questions (5 / 8 Marks)

Nature of Flow & Frictional Pressure Loss (2081 Final, Q1):

Oil is pumped at a rate of 1.9 litres per minute along a horizontal pipe of diameter 0.01 m and length 120 m and has a viscosity of 0.9 Ns/m² and density of 940 kg/m³. Determine the nature of flow. Calculate the pressure lost in the pipe due to frictional effects and calculate the maximum flow rate for laminar conditions to prevail. [8 Marks]

Laminar Velocity Gradient & Wall Shear Stress (2080 Final, Q1):

Oil of specific gravity 0.7 and viscosity 10 poise flows through a pipe of diameter 100 mm. The velocity at the center is 2 m/s. Find: (i) Mean velocity and radial distance at which this occurs. (ii) Pressure gradient in the direction of flow. (iii) Shear stress at the pipe wall. (iv) Reynolds number. [8 Marks]

Laminar Flow Pressure Gradient & Shear Stress (2079 Supp, Q1a):

Oil of density 750 kg/m³ and dynamic viscosity 0.24 poise flows through a 60 mm diameter pipe of length 1000 m at the rate of 0.25 lps. Determine (i) pressure gradient, (ii) wall shear stress. [4 Marks]

Turbulent Flow Velocity Profile Sketch (2079 Supp, Q1b):

Show mathematically how you can relate the centerline and local velocities for a turbulent flow in pipes with supporting sketches. [4 Marks]

Roughness & Power Relationship (2079 Regular, Q1):

A CI water pipe of 0.5 m diameter was originally designed to carry a discharge of 250 lps in a water supply pipeline of Mahendranagar city. After 10 years of use, it was found out that the pipeline required 30% more power to carry the same discharge. Determine the corresponding roughness of pipe and rate of change in roughness if $\text{power} \propto \text{friction factor } f$. Take $\nu_{\text{water}} = 0.014$ stokes. [6+2 = 8 Marks]

Turbulent Velocity Relationship (2079 Regular, Q2b):

Show mathematically how you can relate the maximum and point velocities for a turbulent flow in pipes with supporting sketches. [5 Marks]

Prandtl's Mixing Length Theory (2081 Final, Q4b):

What is Prandtl's mixing length theory? [3 Marks]

Wall Distance for Average Velocity (2082 Final, Q8a):

Determine the distance from the pipe wall at which the local velocity is equal to the average velocity for turbulent flow in pipes. [3 Marks]

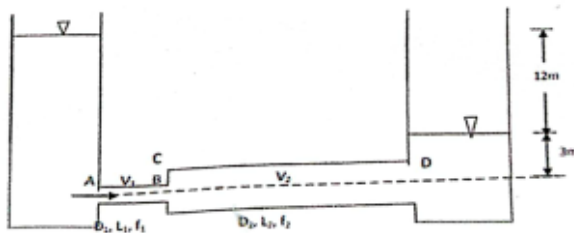
Hydraulic Radius vs. Depth Definition (2082 Final, Q5b):

Define hydraulic radius (R) and hydraulic depth (D) with formulas, and state why they matter in open channel flow. [3 Marks]

Long Answer Questions (10 / 15 Marks)

HGL and TEL for Compound Pipe (2081 Mid-Term, Q3):

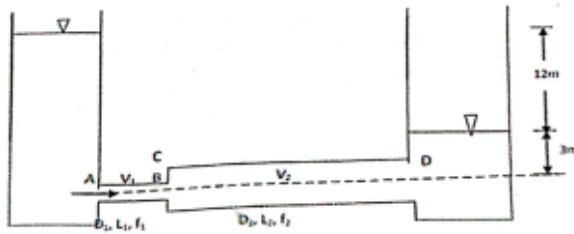
Two reservoirs whose water surface elevation differs by 12 m are connected by the following horizontal compound pipe system starting from the high-level reservoir as shown in the figure. $L_1 = 200$ m, $D_1 = 0.2$ m, $f_1 = 0.008$, $L_2 = 500$ m, $D_2 = 0.4$ m, $f_2 = 0.006$. Compute all the losses of the head. Sketch EGL and HGL lines. [12 Marks]



Compound Pipe Head Loss & HGL/TEL (2080 Final, Q2):

Compute the total loss of head in a reservoir combination whose surface elevations differ by 12 m and are connected by a horizontal compound pipe system starting from the reservoir at higher

level as shown in the figure. $L_1 = 150$ m, $D_1 = 0.35$ m, $f_1 = 0.007$, $L_2 = 600$ m, $D_2 = 0.8$ m, $f_2 = 0.005$. [8 Marks]



TEL & HGL Theory (2082 Final, Q6a):

Differentiate between the Hydraulic Grade Line (HGL) and the Total Energy Line (TEL) for flow through pipes. [2 Marks]

Minor Losses Definition (2082 Final, Q7a):

Write different types of head losses in pipe flow and explain any two of them. [3 Marks]

Unit 2: Pipe Flow Problems

Focuses on pipes in series/parallel, siphons, branching pipes (three-reservoir problem), Hardy Cross network iteration, and power transmission.

Short Answer Questions (5 / 8 Marks)

Parallel Pipe Design for Power Saving (2081 Final, Q2):

Water is pumped through a 600 mm diameter pipe with a head loss of 20 m. It is intended to reduce the power consumption by laying another parallel pipe of appropriate diameter laid by the side of the existing pipeline with the same overall length and the same friction factor - by reducing the head loss to 10 m, but still delivering the same discharge jointly through both pipes. What should be the diameter of this additional pipe? [8 Marks]

Siphon Summit Pressure & Discharge (2079 Supp, Q2):

A 180 mm diameter siphon connects two reservoirs having a difference in elevation of 16 m. The length of the siphon is 450 m and the summit is 4.5 m above the upper reservoir level. The length of the pipe from upper reservoir to the summit is 90 m. Determine the discharge through the siphon and pressure at the summit. Consider all losses. Take $f = 0.025$. [8 Marks]

Equivalent Series Pipe (2079 Regular, Q2a):

Calculate the diameter of a single pipe which can replace a series combination of three pipes of lengths 1000 m, 400 m and 600 m and of diameters 0.5 m, 0.4 m and 0.3 m respectively. [3 Marks]

Siphon Design with Air Separation Limit (2080 Final, Q5):

A siphon of diameter 180 mm connects two reservoirs having a difference in elevation of 18 m. The length of the siphon is 750 m and the summit is 4.5 m above the water level in the upper reservoir. If the separation takes place at 2.5 m of water absolute, find the maximum length of the siphon from upper reservoir to the summit. Take $f = 0.016$ and atmospheric pressure = 10.2 m of water. Neglect minor losses. [8 Marks]

High-Viscosity Pipe Diameter Calculation (2082 Final, Q3b):

A liquid having a specific gravity of 2.8 and a viscosity of 0.8 poise flows through a smooth pipe of unknown diameter. The flow results in a pressure drop of 800 N/m² over a pipe length of 2 km. If the mass flow rate of the liquid is 2500 kg/h, determine the diameter of the pipe. [3 Marks]

Series vs. Parallel Head Loss Ratio (2082 Final, Q4):

Two pipes, each of length L , have diameters D_1 and D_2 , respectively. When the pipes are connected in parallel, a total discharge Q flows through the system and the corresponding head loss is H_1 . When the same two pipes are connected in series, the same total discharge Q flows through the system and the corresponding head loss is H_2 . Assuming that minor losses are negligible and that the Darcy-Weisbach friction factor f is the same for both pipes, determine the ratio of H_1 to H_2 given that $D_2 = D_1/2$. [5 Marks]

Siphon Definition & Conditions (2082 Final, Q5a):

Define a siphon and state the conditions required for continuous flow through it. [2 Marks]

Power Transmission Diameter & Efficiency (2081 Final, Q5):

A pipe line of length 2000 m is used for power transmission. If 110.36 kW power is to be transmitted through the pipe in which water having a pressure of 490.5 N/cm² at inlet is flowing. Find the diameter of the pipe and efficiency of transmission if the pressure drop over the length of pipe is 98.1 N/cm². Take $f = 0.0065$. [8 Marks]

Long Answer Questions (10 / 15 Marks)

Three-Reservoir Problem (2081 Final, Q7):

For the three reservoir problem $Z_1 = 150$ m, $Z_3 = 30$ m, $Q_2 = 0.65$ m³/s and $r_1 = 4.0$, $r_2 = 40$ and $r_3 = 15$ for the three pipes while n in the equation $h_f = rQ^n$ is 2.0. Determine Q_1 , Q_3 and Z_2 . [15 Marks]

Three-Reservoir Problem (2080 Final, Q7):

Reservoir A, water surface elevation 120 m is connected to reservoir B and C having surface elevation 70 m and 50 m respectively. A pipeline 150 mm diameter and 400 m long connects

reservoir A to junction D. Reservoir B and C are connected to junction D by 75 mm diameter 100 m long and 100 mm diameter 250 m long pipeline respectively. Assuming friction factor $f = 0.04$ for all pipes, estimate the rate of flow for each pipe, neglecting minor head losses. **[15 Marks]**

Three-Reservoir Problem (2079 Supp, Q8a):

Reservoir A with water surface elevation 120 m is connected to reservoir B and C having water surface elevations of 70 m and 50 m respectively. A pipeline 180 mm diameter and 450 m long connects reservoir A to junction D. Reservoir B and C are connected to junction D by 80 mm diameter 110 m long and 90 mm diameter 220 m long pipeline respectively. Assuming friction factor $f = 0.035$ for all pipes, estimate the discharge through all pipes. Neglect minor head losses. **[12 Marks]**

Compound Pipe & Siphon (2079 Supp, Q7a):

The water surface of reservoir B is 10 m below the water surface in reservoir A. The reservoirs are connected by horizontal compound pipe system 1 and 2. The pipe details are:

Pipe 1: Length = 450 m, Diameter = 0.35 m, $f = 0.007$

Pipe 2: Length = 250 m, Diameter = 0.25 m, $f = 0.008$

Compute all head losses. (i) Compute Q , (ii) Sketch EGL and HGL. **[7+5 = 12 Marks]**

Sudden Enlargement Pressure Analysis (2079 Regular, Q7a):

A 180 m long pipeline ABC is laid on an upward slope of 1 in 40. The pipe details are:

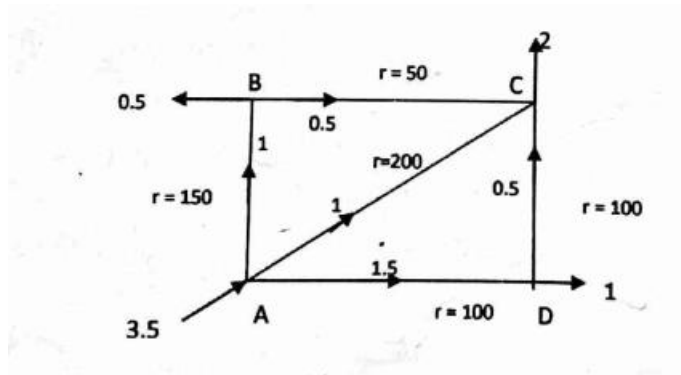
Section AB: Length = 100 m, Diameter = 0.1 m

Section BC: Length = 80 m, Diameter = 0.2 m

The pipe suddenly enlarges at B. A discharge of $0.2 \text{ m}^3/\text{s}$ is pumped into the pipe at its lower end at A and is discharged into a closed tank at C. The pressure at supply end is 200 kN/m^2 . $f = 0.008$. (i) What is the pressure at C? (ii) Draw HGL and EGL. **[6+6 = 12 Marks]**

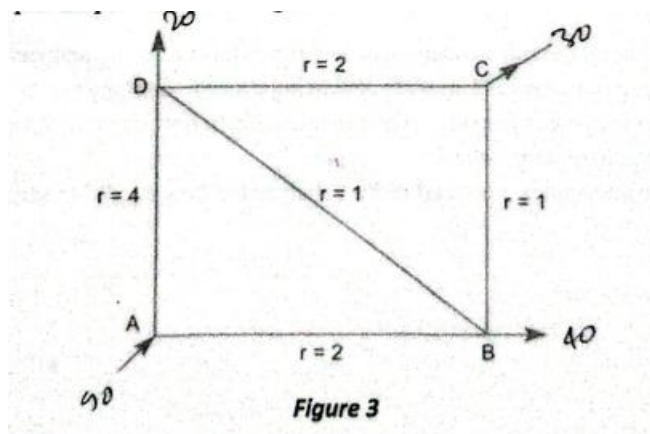
Hardy Cross Pipe Network - Double Loop (2079 Regular, Q8):

A water supply distribution network in Terai is as shown in the figure below. The numbers and arrows indicate initial assumptions for discharge and flow direction for each pipeline respectively. Perform two iterations to calculate the corrected discharge in the pipe network. If its inlet is supplied by a 20 m high overhead tank, calculate pressure at points B, C and D neglecting velocity heads. State necessary assumptions clearly. **[10+5 = 15 Marks]**



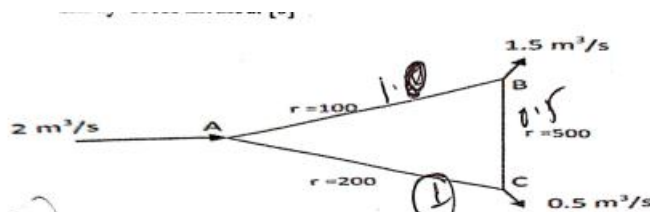
Hardy Cross Pipe Network - Double Loop (2082 Final, Q10b):

Using the Hardy-Cross method, determine the discharge in each pipe of the pipe network shown in Figure 3. The network consists of five interconnected pipes. The head loss in each pipe is given by $h_f = rQ^2$ where Q is the discharge through the pipe and r is the pipe resistance coefficient. The values of r for the individual pipes are provided in the figure. [5 Marks]



Hardy Cross Pipe Network - Single Loop (2081 Mid-Term, Q2):

For the pipe network shown in the figure, obtain the discharge distribution by using the Hardy-Cross method. [8 Marks]



Unit 3: Unsteady Flow in Pipe

Focuses on U-tube oscillations, surge control (surge tanks), and water hammer pressure rise calculations.

Short Answer Questions (5 / 8 Marks)

Water Hammer - Fast vs. Slow Closure & Peak Pressure (2081 Final, Q3):

What is water hammer effect? A cast iron pipe of 350 mm diameter and 9 mm thick is 1500 m long. The pipe is to convey 220 lps of water. Calculate: (i) the maximum time of closure of a valve at the downstream end that would be recognized as rapid closure. (ii) peak water hammer pressure. Take $K = 2 \times 10^9 \text{ N/m}^2$ (bulk modulus of water) and $E = 1 \times 10^{11} \text{ N/m}^2$ (modulus of elasticity of pipe material). **[8 Marks]**

Surge Tank Necessity (2081 Mid-Term, Q1):

What do you understand by a surge tank? Why is it necessary in the operation of a hydroelectric plant? **[5 Marks]**

Pressure Variation in Pipe Centerline (2079 Supp, Q3a):

A steel pipe is connected directly to an oil storage reservoir. The valve provided at the end of the pipe is suddenly closed to stop the flow of oil. Explain with necessary figures the variation of pressure in the mid-section of the steel pipe. **[5 Marks]**

Pressure Variation in Penstock (2079 Regular, Q3a):

A 800 m long penstock pipe supplying water to turbine is connected directly to a reservoir of a storage type hydropower project. The valve connecting to the turbine was forced to close in 1.5 seconds due to sudden damage in the transmission line. Explain with necessary figures the variation of pressure in the penstock pipe at 400 m d/s of the reservoir. Take $c = 1000 \text{ m/s}$. **[5 Marks]**

Water Hammer & Remedial Measures (2080 Final, Q3b):

What is water hammer phenomena? Also point out its effects and remedial measures. **[3 Marks]**

Long Answer Questions (10 / 15 Marks)

Gradual Valve Closure Risk Assessment (2082 Final, Q7b):

A concrete pipeline, 3.2 km long and 0.30 m in diameter, conveys water to a reservoir at a discharge of 9 million L/day. Flow is controlled by a valve located at the downstream (reservoir) end. Under normal operating conditions, the valve is closed gradually over 16 s. The maximum allowable pressure head for the pipe is 25 m of water. (i) Determine the pressure head rise produced by the gradual valve closure and assess whether the pipe is at risk of failure. Justify your answer. (ii) Explain the concept of water hammer. Then, applying this concept, explain why

a sudden closure of the downstream valve does not cause the flow to stop instantaneously along the entire length of the pipeline. [4+3 = 7 Marks]

Unsteady Flow Momentum Equation (2080 Final, Q8a):

Derive the momentum equation for unsteady flow through pipes. [5 Marks]

Water Hammer Definition & Remedies (2079 Regular, Q7b / 2079 Supp, Q7b):

What is water hammer phenomena? Also point out its effects and remedial measures. [1+2 = 3 Marks]

U-Tube Oscillation Theory (2079 Supp, Q11i):

Write short notes on: Oscillation of liquid in a U-tube. [5 Marks]

Surge Control Theory (2080 Final, Q11a):

Write short notes on: Surge control. [3 Marks]

PART B: Fluid Flow in Open Channel (60% Weightage)

Unit 4: Basics of Open Channel Flow

Focuses on geometrical properties of sewer lines (partially filled circular conduits) and types of flow based on Froude number.

Short Answer Questions (5 / 8 Marks)

Sewer Line Geometry - Flow Depth < Radius (2079 Supp, Q8b):

Determine the geometric properties of a sewer line made up of precast RCC pipe with flow depth smaller than radius. [3 Marks]

Sewer Line Geometry - Flow Depth > Radius (2079 Regular, Q3b):

Identify the geometric properties of a sewer line made up of hume pipe with flow depth greater than its radius. [3 Marks]

Froude Number Flow Classification (2081 Mid-Term, Q4a):

Discuss the types of flow based on the Froude number and their distinguishing features. [2 Marks]

GVF and RVF Definitions & Examples (2081 Mid-Term, Q4b):

Define gradually varied flow (GVF) and rapidly varied flow (RVF). Provide examples of each. [2 Marks]

Open Channel Classification (2080 Final, Q10a):

Classify open channel flow. [5 Marks]

Unit 5: Uniform Flow

Focuses on Manning's/Chezy's equations, normal depth, and best hydraulic cross-sections.

Short Answer Questions (5 / 8 Marks)

Most Efficient Trapezoidal Channel Derivation (2081 Final, Q4a):

Derive the expression for a most efficient trapezoidal channel with varying depth and fixed side slopes. [5 Marks]

Manning's n calculation for Trapezoidal Channel (2080 Final, Q4b):

Calculate the Manning's coefficient for a trapezoidal channel with side slopes 1:1 carrying a discharge of $1 \text{ m}^3/\text{s}$ at a depth of 1 m and laid at a slope of 0.001. [3 Marks]

Chezy's C for Triangular Channel (2079 Regular, Q4b):

Calculate the Chezy's coefficient for a triangular channel with an apex angle of 60° carrying a discharge of $1 \text{ m}^3/\text{s}$ at a depth of 0.75 m and laid at a slope of 0.009. [3 Marks]

Trapezoidal Bed Shear Stress & Friction Factor (2079 Supp, Q4):

A trapezoidal channel with bottom width 2 m and side slope 2H:1V, flows with a depth of 0.8 m in a slope of 0.001, compute bed shear stress, shear velocity, Chezy's C and friction factor f . [8 Marks]

Most Economical Half-Hexagonal Section (2079 Supp, Q5):

What do you mean by energy slope? An open channel of the most economic cross section, having the form of half hexagon with horizontal bottom is required to carry a flow of $20.2 \text{ m}^3/\text{s}$ of water. The slope of the channel bottom is 1 in 2500. Taking Manning's $n = 0.015$, determine the dimensions of the cross sections. [8 Marks]

Trapezoidal Channel with Uniform Flow Bed Slopes (2079 Regular, Q5):

A trapezoidal channel 2.5 m wide and having side slopes 1:1 is carrying a discharge of $1.1 \text{ m}^3/\text{s}$ at a specific energy of 0.6 m. What would be the corresponding bed slopes required to sustain the uniform flow if $n = 0.015$? Also calculate the minimum specific energy required to carry this discharge. [8 Marks]

Rectangular vs. Triangular Discharge Ratio (2079 Regular, Q4a):

Find the ratio of discharges in concrete lined rectangular and triangular channels, if the rectangular channel has flow depth equal to its width and the triangular channel has an apex angle of 90° , such that the flow areas are same for both the channels. [5 Marks]

Most-Economical Section Concept (2082 Final, Q3a):

What do you mean by 'Most-economical section' of an open channel? [2 Marks]

Long Answer Questions (10 / 15 Marks)

Lined vs. Unlined Channel Economics (2081 Final, Q8b):

A trapezoidal channel having side slope of 1:1 has to carry a flow of $20 \text{ m}^3/\text{s}$. The bed slope is 1 in 950. Chezy's constant C is 40 if the channel is unlined and 75 if the channel is lined with concrete. The cost per m^3 of excavation is 2.8 times the cost per m^2 of lining. Find which arrangement is economical. [8 Marks]

Lined vs. Unlined Channel Economics (2081 Mid-Term, Q5):

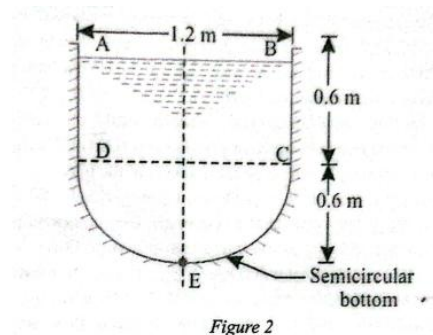
A trapezoidal channel with a side slope 1:1 has to carry a flow of $15 \text{ m}^3/\text{s}$. The bed slope is 1 in 1000. Chezy's C is 45 if the channel is unlined and 70 if the channel is lined with concrete. The cost per m^3 of excavation is 3 times per m^2 of lining. Find which arrangement is economical. [11 Marks]

Maximum Discharge with Fixed Perimeter (2080 Final, Q8b):

A lined rectangular channel with $n = 0.015$ is 4.5 m wide and has a flow depth of 2 m with bed slope 1 in 1500. Keeping the channel section and the perimeter constant, to what extent can the discharge be increased without changing the bed slope? [10 Marks]

Semicircular and Rectangular Composite Section (2082 Final, Q9b):

Find the discharge of water through the channel shown in Figure 2. Take the value of Chezy's constant = 60 and slope of the bed as 1 in 950. [7 Marks]



Unit 6: Principle of Energy and Momentum

Focuses on specific energy, hump/width transitions, specific force, and critical flow conditions.

Short Answer Questions (5 / 8 Marks)

Rectangular Critical Flow Hump Condition (2080 Final, Q4a):

What is the condition for the flow of maximum possible discharge in a rectangular channel for a constant specific energy? [5 Marks]

Froude Number Relation Derivation (2081 Final, Q10b):

If F_n is the Froude no. for normal depth y_n of a rectangular channel, show that:

$$y_c / y_n = F_n^{2/3}.$$

[6 Marks]

Long Answer Questions (10 / 15 Marks)

Floor Rise Criticality & Energy Loss (2081 Final, Q10a):

A wide rectangular channel carries a flow of $2.5 \text{ m}^3/\text{s}/\text{m}$, the depth of flow being 1.5 m. (i) Calculate the minimum rise in floor level to produce critical flow condition at a section. (ii) What is the corresponding fall in water level? (iii) If the loss of energy over the rise is 15% of u/s velocity head, what will be the minimum rise in the floor to produce a critical flow? [9 Marks]

Rectangular Alternate / Conjugate Depths (2079 Supp, Q10a):

Water flows through a rectangular channel 3 m wide, with a velocity of 1.2 m/s. The depth of flow is 0.5 m. Compute (i) specific energy, (ii) specific force, (iii) alternate depth, (iv) conjugate depth. [8 Marks]

Channel Transition Width Contraction (2079 Supp, Q10b):

A rectangular channel contracts from a width of 3 m to 2.5 m in a short smooth transition section. If the discharge is $6.5 \text{ m}^3/\text{s}$, and the u/s depth is 0.4 m, determine the d/s depth. [7 Marks]

Hump Design and Water Depth Rise (2080 Final, Q10b):

In a 3.5 m wide rectangular channel laid at a slope of 1 in 3300, uniform flow occurs at a flow depth of 2 m. Calculate how high a hump can be raised on the channel bed without causing a change in the u/s flow depth. If the u/s flow depth is to be raised to 2.4 m, what should be the height of hump? Take $n = 0.014$. [10 Marks]

Hump vs. Width Reduction (2079 Regular, Q10):

A discharge of $2.5 \text{ m}^3/\text{s}$ flows through a 2.2 m wide rectangular channel at a depth of 1 m. For the following changes in the channel section, check if critical flow occurs at a section with:

a) Constriction in channel with width reduction to 1.5 m.

b) Introduction of a smooth hump 0.2 m high.

c) Combination of hump and width constriction stated above.

Will case (c) cause increase of u/s depth? If so, to what extent? [4+4+4+3 = 15 Marks]

Channel Transition Energy Calculations (2082 Final, Q8b):

An 8 m wide channel conveys 15 m³/s of water at a depth of 1.2 m. Calculate: (i) Specific energy of the flowing water, (ii) Critical depth and critical velocity, and (iii) Froude number and state the nature of flow. [7 Marks]

Unit 7: Gradually Varied Flow (GVF)

Focuses on bed slope classification, surface profile plots, and backwater profile Direct Step computations.

Short Answer Questions (5 / 8 Marks)

GVF Bed Slope Classification (2082 Final, Q1):

Explain the types of bed slope in gradually varied flow with their characteristics. [5 Marks]

GVF Backwater Surface Examination (2082 Final, Q2):

Water flows at a rate of 7.5 m³/s in a 2 m wide rectangular channel having bottom slope = 0.009. If a dam at d/s increases the depth to 4 m, examine the water surface profile u/s from the dam. Take $n = 0.015$. [5 Marks]

GVF Grade Transitions & Sluice Gate Plots (2081 Final, Q6):

Plot the possible types of GVF profiles for the change in grades as:

(i) Mild - Steep - Mild.

(ii) Mild - Steep slope with a sluice gate in the mild slope.

Also explain in detail about the S₁ type flow profile. [8 Marks]

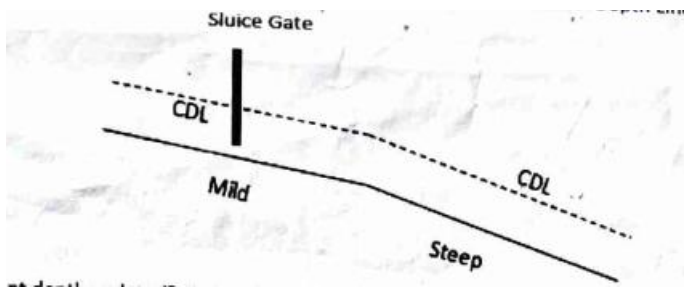
GVF Profile Change Plot - Steep-Mild-Steep (2080 Final, Q6a):

Plot the possible types of GVF profiles for change in grades as 'Steep - Mild - Steep'. [3 Marks]

GVF Profile Change Plot - Sluice Gate in Mild-Steep (2079 Supp, Q6a / 2079 Regular, Q6a):

Plot the possible types of GVF profiles for given arrangement, CDL = Critical Depth Line, NDL = Normal Depth Line.

(i) Sluice Gate in Mild slope followed by Steep slope. [3 Marks]



Long Answer Questions (10 / 15 Marks)

Trapezoidal Channel 3-Step Backwater Curve (2080 Final, Q9):

A trapezoidal channel having a bottom width of 6 m, side slope 2H:1V, $n = 0.025$, and $s = 0.0016$ carries a discharge of $10 \text{ m}^3/\text{s}$. Compute the backwater profile created by a dam which backs up the water to a depth of 2.0 m immediately behind the dam. Use direct step method for computation and solve in 3 steps. [15 Marks]

Sluice Gate Vena Contracta to Jump Distance (2079 Supp, Q9):

A 5 m wide long rectangular channel has a vertical sluice gate with an opening of 0.7 m producing a downstream jet depth of 0.38 m supplying a steady discharge of $20 \text{ m}^3/\text{s}$. Assuming that the flow downstream of the gate eventually returns to the uniform flow depth of 2.5 m, $n = 0.02$, $S_0 = 0.01$.

- Verify that hydraulic jump occurs.
- What distance d/s of the vena contracta does the hydraulic jump occur? Use direct step method and solve in two steps.
- Will the hydraulic jump be affected in any ways if the uniform flow depth changes to 3 m?

[4+8+3 = 15 Marks]

Vena Contracta to Jump Distance (2081 Final, Q9):

Water flows from an under sluice into a very wide rectangular channel. The channel has a bed slope of 1 in 1000. The sluice is regulated to discharge $6 \text{ m}^3/\text{s}$ per m width of channel, the depth at vena contracta being 0.5 m. Will a hydraulic jump form? If so, determine its location in two steps using direct step method. Take $n = 0.015$. [15 Marks]

Unit 8: Rapidly Varied Flow (RVF)

Focuses on hydraulic jumps, energy loss percentage, sequent depth relations, and Froude number classifications.

Short Answer Questions (5 / 8 Marks)

Hydraulic Jump Sequent Depth Derivation (2079 Regular, Q6b / 2079 Supp, Q6b):

How are sequent depths related? Derive the relation with necessary assumptions. [5 Marks]

Hydraulic Jump Sequent Depth Diagrams (2080 Final, Q6b):

Show the relation between sequent depths in a hydraulic jump using specific energy diagram and specific force diagram. [5 Marks]

Hydraulic Jump Energy Loss Percentage (2080 Final, Q3a):

What percentage of energy is lost in a hydraulic jump when $Fr_1 = 9$ and $y_1 = 0.4$ m? [5 Marks]

Long Answer Questions (10 / 15 Marks)

Sluice Gate Jump Verification & Jet Depth Rise (2082 Final, Q6b):

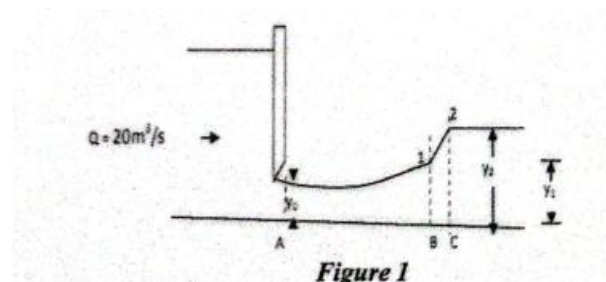
A vertical sluice gate with an opening of 0.67 m produces a downstream jet depth of 0.4 m when installed in a long rectangular channel 5 m wide conveying a steady discharge of $20 \text{ m}^3/\text{s}$.

Assuming that the flow downstream of the gate eventually returns to the uniform flow depth of 2.5 m. (Refer to Figure 1)

(i) Verify that a hydraulic jump occurs.

(ii) If the downstream depth is increased to 3 m, analyze the flow condition at the gate.

[8 Marks]



3. Analysis of High-Yield Repeated Questions

Hydraulics exhibits the highest rate of direct problem repetition of any fourth-semester civil engineering subject. Mastering these specific numerical templates will secure the vast majority of exam marks:

1. The "Water Flowing from Under Sluice to Jump" GVF Numerical

Repetition Cycle: 2081 Final (Q9) and 2079 Supp (Q9) are almost identical. Both require verifying a hydraulic jump and then using the Direct Step Method (in 2 steps) to find the distance from the vena contracta to the jump location.

Slight variations:

2081 Final Q9: Wide rectangular channel, $q = 6 \text{ m}^3/\text{s}/\text{m}$, $y_1 = 0.5 \text{ m}$, $S_0 = 0.001$, $n = 0.015$.

2079 Supp Q9: Channel width = 5 m, $Q = 20 \text{ m}^3/\text{s}$ ($q = 4 \text{ m}^3/\text{s}/\text{m}$), $y_1 = 0.38 \text{ m}$, $S_0 = 0.01$, $n = 0.02$.

2. The Siphon Summit Pressure & Discharge Problem

Repetition Cycle: 2080 Final (Q5) and 2079 Supp (Q2) are direct structural matches.

Formulas Needed:

Apply Bernoulli's between reservoir surface and summit to find pressure head (p_s/γ).

Check the cavitation separation limit (2.5 m absolute vs. gauge).

3. Symmetrical Composite Channel Flow (Semicircular Bottom + Rectangular Top)

Repetition Cycle: 2082 Final (Q9b) is a classic custom channel shape.

How to solve: Splitting the channel into Area 1 (semicircle of diameter 1.2 m) and Area 2 (rectangle $1.2 \text{ m} \times 0.6 \text{ m}$), summing the wetted perimeters ($P = \pi R + 2 \times 0.6$), and applying Chezy's:

$$Q = A C \sqrt{(R_h S_0)}$$

4. Hardy-Cross Pipe Networks

Repetition Cycle: This is a guaranteed 10 to 15-mark question in every paper.

Double loops with diagonal pipes: 2082 Final (Q10b) and 2079 Regular (Q8).

Single triangular loops: 2081 Mid-Term (Q2).

Step-by-step procedure: Set up table with columns:

Pipe, r , Q_0 , $h_f = rQ|Q|$, $2r|Q|$

Calculate correction:

$$\Delta = -\Sigma h_f / \Sigma (2r|Q|)$$

Apply Δ to close loops.

4. Subject Preparation Priorities & Strategy

Based on past-year weightages and syllabus hours, here is your chronological study planner, ranked from highest return-on-investment (ROI) to lowest:

Priority	Unit Name	Approx. Mark %	Primary Areas of Focus
1 (Highest)	Unit 2: Pipe Flow Problems	20% - 25%	Hardy Cross loop iterations (Double loops with diagonals are highly probable), 3-reservoir branching pipes, Siphon discharge & pressure.
2 (Highest)	Unit 7: Gradually Varied Flow	18% - 22%	Direct Step Method (2-step and 3-step tables), plotting profile curves for grade transitions (Steep-Mild, Mild-Steep with sluice gate).
3 (High)	Unit 6: Energy & Momentum	15% - 18%	Rectangular channel hump heights, width transitions (solving for downstream critical and subcritical depths).
4 (High)	Unit 8: Rapidly Varied Flow	12% - 15%	Hydraulic jump conjugate/sequent depth ratio derivations, energy loss percentage calculations, applications of jumps.
5 (Medium)	Unit 5: Uniform Flow	10% - 12%	Best hydraulic cross-section dimensions (especially half-hexagon and trapezoidal), Composite semicircular-rectangular channel discharges.
6 (Medium)	Unit 1: Basics in Pipe Flow	8% - 10%	Laminar velocity distributions, wall shear stress, Reynolds number, compound series pipe diameters.
7 (Low)	Unit 3: Unsteady Flow	5% - 8%	Water hammer rapid vs. gradual closure calculations, surge tank theory.
8 (Lowest)	Unit 4: Basics of Open Channel	2% - 5%	Hume pipe/sewer line geometric property formulas (depth > radius vs. depth < radius).